

Literature Review Section

Review of UWB/RTLS Workforce Tracking and Smart Attendance Systems

Item	Details
Project phase	Pre-Implementation Phase – Literature Review
Research question addressed	RQ1 – What UWB/RTLS and smart-attendance approaches exist for tracking workforce movement in a facility, and what accuracy, coverage and limitations do they report? How do they compare with other indoor positioning options?
Related project objectives	O1 (real-time movement tracking), O2 (visited locations), O3 (time spent at locations)
Version / date	v1.0 draft – 10/10/2026

1. Introduction and Method

EgySpy aims to give management continuous visibility of employee movement inside a defined facility, beyond a single attendance record. This section reviews systems that track or verify the presence of people: UWB and other real-time location systems (RTLS), and smart attendance systems. UWB/RTLS is the main focus; other indoor positioning options are covered at comparison level, so technology stays open as the proposal requires.

Sixteen sources were identified through targeted searches using the strings in the review plan, and their bibliographic details were checked against publisher and indexing records: four surveys [1, 2, 3, 4], six experimental evaluations of UWB [5, 6, 7, 8, 9, 10], three workforce-tracking applications [11, 12, 13] and three smart attendance systems [14, 15, 16]. Few sources study UWB tracking of office staff, so evidence from construction, industrial and care settings is used and treated as transferred.

2. How UWB Tracking Works

UWB systems measure ranges between a mobile tag and fixed anchors at known positions, then compute the tag position [10]; common principles include angle of arrival, time of flight and signal strength [2]. Very short UWB pulses allow precise ranging and can penetrate obstacles [3]. Accuracy still depends on the number, layout and surveyed position of the anchors [10], and non-line-of-sight (NLOS) conditions are a recurring error source addressed through error correction and sensor fusion [3]. Further comparisons of UWB systems and in-market modules are given in [1, 4].

3. Comparison of Indoor Positioning Technologies

Table 1 – Technologies as reported in the reviewed sources (figures depend on each study's conditions)

Technology	Reported accuracy / capability	Limitations noted in the sources
UWB	Centimetre-level in favourable conditions, far better than BLE and Wi-Fi [3]. Average error 0.3–0.6 m for a commercial system in an industrial laboratory [7].	Needs fixed anchors, site survey and calibration [9]; sensitive to anchor layout [10] and NLOS [3]. Measured accuracy of two commercial systems was worse than declared [6].
BLE	About 5 m with signal-strength methods [3]; angle-of-arrival finding narrows the gap to UWB [7].	Regarded as a proximity technology compared with UWB [7].
Wi-Fi	5–15 m with signal-strength methods; Wi-Fi round-trip-time is an emerging higher-precision option [3].	Metre-level accuracy with signal-strength methods [3].

Technology	Reported accuracy / capability	Limitations noted in the sources
RFID	Presence detection at reader points; UWB RTLS reported more accurate than traditional RFID-based sensing on jobsites [9].	A registered card passes even if held by another person; read range is limited [16].

UWB is the strongest candidate for accurate, continuous tracking; BLE, Wi-Fi and RFID fit presence checks or coarse zones. The technology decision is left to the architecture analysis.

4. UWB/RTLS Systems and Studies

Table 2 – Evaluations of UWB positioning systems

Ref.	System / study and setting	Reported results	Strengths	Limitations
[5]	Ubisense, BeSpoon and DecaWave compared under identical conditions in an industrial environment, with line-of-sight and NLOS.	DecaWave outperformed BeSpoon, which outperformed Ubisense.	Fair comparison under common conditions.	Single environment; hardware has since evolved.
[6]	Pozyx and DecaWave DW1000, static and dynamic tests.	Both performed below expectations and worse than declared.	Includes dynamic movement.	Two systems; robot-driven reference path, not a person.
[7]	Pozyx UWB in an industrial laboratory; static tests at three heights (including worker-helmet height) and a dynamic robot test.	Static average error 0.3–0.6 m, higher at low heights; dynamic 0.3–0.4 m; judged acceptable for tracking people or objects.	Heights relevant to people; many ground-truth points.	Robot at 0.3 m height; one product.
[8]	UWB system tracking a person moving indoors, against motion-capture ground truth.	Static and dynamic accuracy characterised for 3-D trajectories.	Millimetre-accuracy ground truth; person-oriented.	One system, controlled setting.
[9]	Commercial UWB RTLS on a large construction jobsite.	Accuracy significantly affected by deployment choices (2-D vs 3-D, fewer tags, wired setup, multi-direction coverage).	Large-scale test; reports deployment effort.	Wired anchors add effort and safety implications; not an office setting.

Table 3 – UWB applications for tracking people

Ref.	System / study and setting	Reported results	Strengths	Limitations
[11]	Low-cost UWB proximity warning prototype for construction workers; works with or without fixed anchors.	Designed around cost, site conditions and data protection; GDPR-compliant.	Privacy in the design; usable without fixed infrastructure.	Collision-hazard proximity only, no location or time records; prototype.
[12]	UWB proximity detection with alarms on a worker-worn device and hazard-zone unit; can stop heavy equipment.	Validated in the laboratory and in the field (construction site, steel factory).	Field validation; integrated alerting.	Safety-oriented; needs worn and equipment-side units; not for attendance.
[13]	UWB/BLE tracking for older adults at home; tags trade accuracy against energy.	Room-level accuracy suffices for occupancy; sub-metre needed for wandering detection.	Links use to accuracy requirement; energy-adaptive tags.	Tag charging was a serious problem for users; environment strongly affects accuracy.

The workforce applications reviewed are built for safety, detecting proximity or entry into danger zones [9, 11, 12]. They show that tracking workers with UWB is feasible on real sites, but none records visited locations or time spent as an accountability record.

5. Smart Attendance Systems

Table 4 – Smart attendance systems

Ref.	System	How presence is established	Strengths	Limitations for movement monitoring
[14]	Cloud IoT attendance system (web server, Raspberry Pi 4, temperature sensor) combining facial recognition and QR code.	Check-in verified by face and QR code.	Compares RFID, fingerprint, QR and face approaches; tested in real-time environments.	Check-in event only; no movement tracking described.
[15]	Mobile-device attendance limited to a place using Wi-Fi network localization.	Presence tied to a Wi-Fi network or area.	Reuses existing Wi-Fi.	Area-level only; Wi-Fi localization is coarse (about 5–15 m) [3].
[16]	Raspberry Pi with camera face recognition plus RFID reader, for students, staff and industry employees.	Entry confirmed by card and face.	Two-factor identity check; configurable time window.	RFID alone identifies the card, not the holder; limited read range; entry point only.

These systems establish presence at a single point in time. Combining RFID with face recognition strengthens identity verification [16], but none describes continuous visibility of location after check-in, which is the situation in the EgySpy problem statement.

6. Key Findings and Implications

1. **Use measured, not declared, accuracy.** Surveys report centimetre-level UWB accuracy [3], but tests show 0.3–0.6 m averages [7] and accuracy below declared figures [6].
2. **Accuracy depends on deployment and environment** (anchor layout [10], deployment choices [9], NLOS [5]). Survey and calibration effort must be planned.
3. **Required accuracy depends on use.** Room-level accuracy suffices for occupancy [13]; recording visited locations and time spent (O2, O3) may therefore be specified at zone level.
4. **Worn tags and privacy are design topics.** Tag charging was a serious issue for users [13], and a UWB workforce prototype was designed for GDPR compliance [11].
5. **Existing solutions are split.** Workforce RTLS focuses on safety [9, 11, 12] and attendance systems are event-based [14, 15, 16]. None of the reviewed sources combines continuous in-facility tracking with attendance accountability; this will be tested in the comparative synthesis.

Limitations of this review. Evidence on UWB tracking of office staff is scarce, accuracy figures come from different test set-ups, several evaluated systems are older hardware, and vendor documentation was not included or verified.

7. References

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